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# Signal representation and load transfer in bearing fault classification

# Representación de señal y transferencia entre cargas en fallas de rodamientos

## Abstract

Comparisons of signal representations for bearing fault diagnosis are usually reported under a single operating condition, where accuracy saturates. This work measures how four representations behave when the test load differs from the training load. Time-domain statistics, FFT magnitude, db4 wavelet band energies and STFT spectrogram were combined with three vector classifiers over the sixteen (training load, test load) pairs of the CWRU drive-end dataset at 12 kHz, with the split made by source record and purged contiguous blocks. Within a single load, where the split shares records, all four exceeded 0.95 macro-F1 on average. Across loads the mean drop ranged from 0.074 for the spectrogram to 0.153 for time-domain statistics, and the latter depended on transfers involving 0 HP. With default hyperparameters the omnibus Friedman test did not reject, which does not show that the representations are equivalent, while the three per-classifier tests each rejected and disagreed on the ranking. The representation by classifier interaction persisted when each load was left out and under nested tuning on training data, which also made the omnibus test reject, so rankings depend on the configuration. With four loads and one bearing per fault condition these inferences are indicative and specific to this dataset. Per-window RMS normalisation gave no clear improvement in the four combinations tested and degraded transfer with Random Forest.

***Keywords***: bearing fault diagnosis, data leakage, domain shift, Friedman test, signal representation, vibration analysis

## Resumen

Las comparaciones entre representaciones de señal para el diagnóstico de fallas en rodamientos suelen reportarse bajo una sola condición de operación, donde la exactitud satura. Este trabajo mide cómo se comportan cuatro representaciones cuando la carga de prueba difiere de la de entrenamiento. Se combinaron estadísticos temporales, magnitud FFT, energías de banda wavelet db4 y espectrograma STFT con tres clasificadores de vector sobre los dieciséis pares de cargas del conjunto CWRU a 12 kHz, con división por registro y por bloques contiguos purgados. Dentro de una misma carga, donde la partición comparte registros, las cuatro superaron en promedio 0,95 de F1-macro. Entre cargas la caída media osciló entre 0,074 para el espectrograma y 0,153 para los estadísticos, y esta última dependió de las transferencias con 0 HP. Con hiperparámetros por defecto la prueba de Friedman ómnibus no rechazó, lo que no demuestra su equivalencia, y las tres pruebas por clasificador rechazaron y discreparon en el orden. La interacción representación-clasificador persistió al excluir cada carga y con un ajuste anidado sobre datos de entrenamiento, que hizo rechazar la prueba ómnibus, por lo que el orden depende de la configuración. Con cuatro cargas y un rodamiento por condición de falla, estas inferencias son indicativas y propias de este conjunto. La normalización RMS no mejoró claramente ninguna de las cuatro combinaciones probadas y degradó la transferencia con Random Forest.

***Palabras clave***: análisis de vibraciones, desplazamiento de dominio, diagnóstico de fallas en rodamientos, fuga de datos, prueba de Friedman, representación de señal

## 1. Introduction

Rolling element bearings fail in ways that leave a signature in the vibration measured on the housing: a localised defect produces an impact each time a rolling element passes over it, exciting the structural resonance at a rate set by the geometry and the shaft speed, which is the basis of envelope analysis [1].

Data-driven diagnosis converts the signal into a representation and trains a classifier on it, and the representation is normally chosen by comparing candidates on the same data. The Case Western Reserve University (CWRU) Bearing Data Center dataset [2] is the standard reference for those experiments, with a benchmark study devoted to it [3] and later work refining that practice [4].

Accuracy within a single operating condition saturates near unity, and the way the data are partitioned decides the result: bearing vibration is usually cut into overlapping windows, and if those are shuffled before splitting, windows sharing samples fall on both sides and the classifier recognises the recording rather than the fault. Data leakage was found in 294 papers across seventeen scientific fields, in some cases producing conclusions far more optimistic than the data supported [5].

The shift between operating conditions is itself an active subject. The CWRU dataset is a common validation reference [6], generalisation remains an open problem across nine datasets [7], and the usual response is a domain-adaptation model aligning a labelled source condition and an unlabelled target condition, for instance with a multi-kernel maximum mean discrepancy [8]. That line fixes a representation and modifies the model, leaving open what the representation itself contributes to the transfer, which is the question addressed here.

Fault diagnosis in rotating machinery has been addressed repeatedly in this journal from measurement rather than model design: time-domain descriptors fed a feed-forward network to diagnose spark-ignition engines [9]; temperature and injection pressure served as measured indicators analysed by multifactorial ANOVA [10]; two measurement methods for stator lamination faults were compared without a learned model [11]; and a four-level Daubechies db4 decomposition was used to estimate fault distance [12], the family used below with five levels.

This study measures how four representations behave when the test load differs from the training load. The contributions of this work are: (1) a record-level evaluation protocol for the CWRU dataset in which the split is performed by source record and, within a record, by purged contiguous blocks, verified by a leakage check with a negative control; (2) a 4×4 matrix of macro-F1 by (training load, test load) pair for four representations and three classifiers, reported per fold; (3) evidence, specific to this dataset and to the configurations evaluated, that the ranking of representations depends on the classifier, with a representation by classifier interaction that persists when any load is left out and under nested tuning; and (4) an ablation, specified before the experiment was run, in which per-window RMS division did not clearly improve cross-load transfer with the two classifiers and representations tested.

## 2. Materials and Methods

### 2.1. Dataset and label verification

Forty drive-end accelerometer records of the CWRU Bearing Data Center [2], sampled at 12 kHz, cover ten class-diameter combinations under four loads (0, 1, 2 and 3 HP, nominally 1797, 1772, 1750 and 1730 rpm). The classes are healthy bearing (Normal), inner race fault (IR), ball fault (B) and outer race fault (OR), the last three at diameters of 0.007, 0.014 and 0.021 inches.

The file naming convention is a known source of error: `99.mat` (Normal, 2 HP) also holds a duplicate of the 1 HP signal, so the channel is matched exactly against the file number (`X{NNN}_DE_time`).

Labels were verified against the physics of the SKF 6205 drive-end bearing rather than third-party tables, by envelope analysis of the 2–5.5 kHz resonance at its characteristic orders; the rolling-element value of 4.7135 tabulated by CWRU corresponds to 2×BSF, not to BSF. The inner race labels were confirmed in 12 of 12 records, the outer race labels in 9 of 12 and the ball labels, by the 2×BSF sideband criterion, in 11 of 12, with weak signatures in the outer race records at 0.014 in and the ball records at 0.021 in; the details are in the supplementary material. All 40 records were kept, since discarding those with a weak signature would bias the result upwards, and fault diameter is a covariate for subgroup reporting.

### 2.2. Windowing and the partition rule

Each record was segmented into windows of 2048 samples with 50 % overlap, yielding 5887 windows. The split is performed by source record, never by window, because overlapping windows of a record share samples.

Outside the diagonal this holds by construction, since training on load $i$ and testing on load $j$ uses different files; on the diagonal it cannot, because the dataset provides a single healthy record per load. One rule was therefore adopted for all sixteen cells: the windows of each record are cut into five contiguous blocks in time and the first window of each internal block is discarded. With a stride of 1024 samples and a window of 2048, windows $m$ and $m + 2$ share no samples, so

one discarded window per boundary suffices, removing 160 windows (2.7 %) and leaving 5727. For cell $(i, j)$ and fold $k$, training uses the blocks other than $k$ from every record of load $i$, and testing block $k$ from every record of load $j$.

On the diagonal, overlap is removed but training and test windows still come from the same record, so the diagonal is a reference and is excluded from every test.

All 80 cell-fold combinations were checked for training windows sharing samples with a test window of the same record, and none was found. The same check without the purge reports 320 violations, which is the 160 overlapping pairs created by four block boundaries in each of the forty records, each counted in the two folds it affects. Table 1 gives the usable window counts; the healthy records last 20 to 40 s against about 10 s for the fault records.

**Table 1.** Usable windows of 2048 samples with 50 % overlap, by class and load, after discarding the boundary windows between blocks.

| Class | 0 HP | 1 HP | 2 HP | 3 HP |
|---|---|---|---|---|
| Normal | 233 | 467 | 468 | 469 |
| IR | 340 | 340 | 340 | 342 |
| B | 341 | 340 | 341 | 341 |
| OR | 341 | 342 | 340 | 342 |

### 2.3. Signal representations

Four representations were computed from the same windows. Figure 1 shows one raw window per class and Figure 2 shows the same inner-race window under the four representations.

**Time-domain statistics.** Twelve classical vibration descriptors, from mean, RMS and kurtosis to the crest, shape, impulse and clearance factors.

**FFT magnitude.** The real FFT magnitude up to Nyquist, 1025 bins of 5.86 Hz.

**Wavelet db4.** Five-level discrete decomposition with the Daubechies db4 wavelet, the family used in [12], as the relative energy of each band (six values: the approximation A5 and the details D5 to D1).

**STFT spectrogram.** The spectrogram in dB, normalised per image to $[0, 1]$, with *nperseg* $= 64$: 187.5 Hz per band, 63 frames and a hop of 2.67 ms. A fault excites the 2–5.5 kHz structural resonance, and its characteristic frequency appears as the repetition rate of the impacts, every 9.3 ms (BPFO) and 6.2 ms (BPFI) at 1797 rpm, so the hop places 2.3 to 3.5 frames within each impact period. The comparison uses a $16 \times 64$ grid of 1024 values, which keeps the hop and halves the frequency axis to 375 Hz; a square $32 \times 32$ grid would leave fewer than two columns per BPFI period. A $64 \times 64$ image of the same transform feeds the convolutional network.

Table 2 summarises the four. The scaler is fitted inside the pipeline, on the training fold only.

**Table 2.** Dimension, effective resolution and dependence on absolute amplitude of each representation.

| Representation | Dim. | Effective resolution | Amp.-inv. | Enters comparison |
|---|---|---|---|---|
| Time-domain statistics | 12 | — | no | yes |
| FFT magnitude | 1025 | 5.86 Hz | no | yes |
| Wavelet db4 (rel. energy) | 6 | dyadic band | yes | yes |
| STFT spectrogram ($16 \times 64$) | 1024 | 375.0 Hz / 2.67 ms | yes | yes |
| STFT spectrogram ($64 \times 64$) | 4096 | 187.5 Hz / 2.67 ms | yes | no |

Two of the four discard amplitude by construction, the wavelet through relative band energy and the spectrogram through per-image normalisation, while the other two keep the absolute scale, which varies between loads. An ablation was therefore fixed before the experiment was run: time-domain statistics and FFT magnitude recomputed on windows divided by their RMS value, with SVM-RBF and Random Forest over the twelve off-diagonal pairs.

### 2.4. Classifiers

Three classifiers that accept vectors were used with all four representations: SVM with RBF kernel, Random Forest with 300 trees and a two-layer perceptron of 128 and 64 units trained for at most 500 iterations, all from scikit-learn [13]. Class weighting was balanced for the SVM and the forest; the perceptron has no class-weight option and was trained unweighted. A standard scaler was fitted on the training fold for the SVM and the perceptron, and omitted for the forest. Otherwise, hyperparameters were left at the library defaults in the protocol, so that all four representations meet each classifier under identical settings. For the SVM this fixes $C = 1$ and $\gamma = 1/(d\,\mathrm{Var}(X))$, which runs from 1/1025 for FFT magnitude to 1/6 for the wavelet energies.

As a sensitivity analysis added in revision, each classifier was also tuned by nested search on training data only: for each training load and fold, each of the four training blocks was held out in turn and the configuration with the best mean inner macro-F1 was refitted on all four, without the test load or block. The 12 SVM, 6 forest and 6 perceptron configurations, the tie rule and the choice in each fold are in the supplementary material. Since the inner validation stays within one load, where macro-F1 saturates, the search optimises in-domain performance, not transfer.

A two-dimensional convolutional network provides a reference row: three blocks of 32, 64 and 128 channels with batch normalisation and max pooling, then global average pooling and a linear layer, in PyTorch [14], trained on $64 \times 64$ spectrograms for a fixed 30 epochs with Adam at $10^{-3}$ and a batch of 64. The epoch count was not tuned, and the network is a descriptive reference only: it has no counterpart for the other three

representations, and an unbalanced design invalidates both Friedman and the ANOVA.

### 2.5. Experimental design and statistical analysis

The design covers 4 representations × 3 vector classifiers × 16 cells × 5 folds, that is 960 fits, plus 80 for the convolutional reference and 240 for the ablation. The seed was fixed at 42 and results stored per fold.

Analyses differ in status. The protocol, fixed on 20 August 2026 before the data were split, specified a Friedman test over the four representations with the off-diagonal pairs as blocks, a Nemenyi post-hoc test and a two-way ANOVA of representation by load distance. On 22–23 August, before the experiment, the primary test was set to macro-F1 averaged over classifiers and folds ($N = 12$, $k = 4$) [15], one Friedman test per classifier was added at a Bonferroni-corrected $\alpha = 0.0167$, and the amplitude ablation was fixed. Only these are confirmatory; every later analysis is exploratory, with uncorrected $p$-values.

Averaging the classifiers keeps $N$ at 12, since counting each (pair, classifier) as a block would rest on their shared partition. The pairs are not independent either: each load appears in six, and $(i, j)$ and $(j, i)$ reuse the same records. The effective $N$ is below 12, so the Friedman test is anticonservative and the Nemenyi critical distance too short, which weakens the rejecting per-classifier tests more than the non-rejecting primary one. Two analyses added in revision address this dependence. A linear mixed model on the cell means models it, with training load and test load as crossed random effects and their interactions with representation and with representation by classifier, tested by likelihood ratio with a 999-replicate parametric bootstrap. A leave-one-load-out jackknife of the Friedman tests, the interaction and the drops measures how much each result depends on a single load. With four loads no inferential procedure is fully reliable, so all $p$-values are indicative.

The two-way ANOVA of the protocol treats folds as replicates although they share four fifths of their training windows, so its degrees of freedom, $p$-values and effect sizes are not interpreted. It was refitted on the 48 cell means, and a three-way model on the 144 (representation, classifier, pair) cell means adds the classifier, with a variant blocking on the pair. These cell means are not independent either, so the residual degrees of freedom are overstated and the $p$-values of these models are anticonservative. The ablation is reported, as in the protocol, as a paired difference over the 60 fold-level differences with a bootstrap interval and Cliff's delta; since folds share training windows, the intervals were also computed over the twelve pairs (supplementary material).

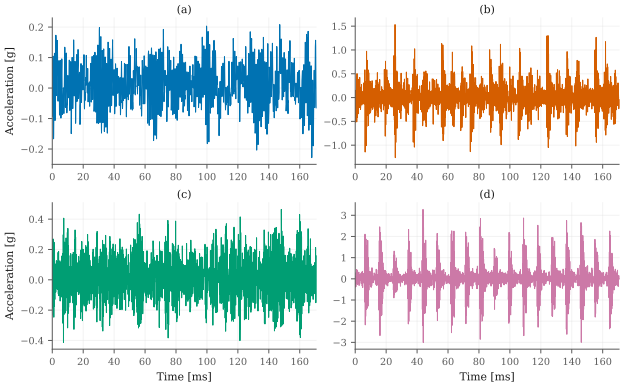



**Figure 1.** Raw drive-end acceleration at 0 HP, one 2048-sample window per class: (a) Normal, (b) inner race, (c) ball, (d) outer race. Each panel has its own $y$-axis scale.

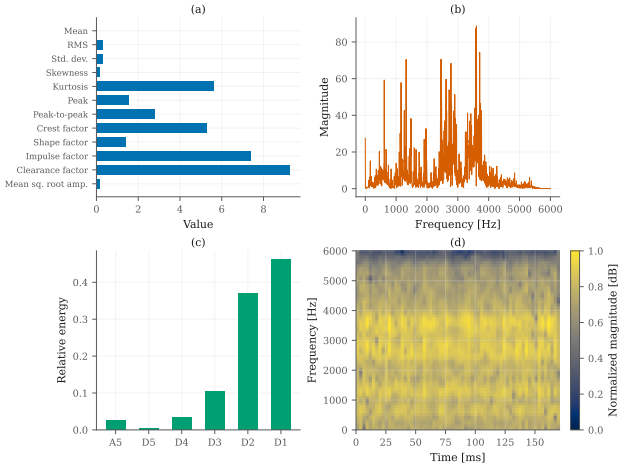



**Figure 2.** Same inner-race window at 0 HP under the four representations: (a) time-domain statistics, (b) FFT magnitude, (c) db4 wavelet relative energy per band, from A5 to D1, (d) STFT spectrogram on the $64 \times 64$ CNN grid, not the $16 \times 64$ grid that enters the comparison.

## 3. Results and Discussion

### 3.1. In-domain performance saturates

Within a single load all four representations exceed 0.95 macro-F1, averaged over classifiers: 0.9997 for FFT magnitude, 0.9925 for the spectrogram, 0.9882 for time-domain statistics and 0.9602 for the wavelet, with the breakdown by classifier in Table 3. Under four hundredths separate best from worst and ten of the twelve representation-classifier combinations sit at or above 0.98, so on this dataset a comparison restricted to the training condition separates little.

**Table 3.** Macro-F1 (mean ± standard deviation over folds and cells) in-domain and out-of-domain, by representation and classifier, with default hyperparameters. Bold marks the best value per column among the twelve compared rows; the in-domain column has two tied at 1.000. The convolutional network, below the line, is a descriptive reference and enters no comparison.

| Representation | Classifier | F1 in-domain | F1 out-of-domain |
|---|---|---|---|
| Statistical | MLP | 0.988 ± 0.015 | 0.799 ± 0.229 |
| Statistical | Random Forest | 0.992 ± 0.008 | 0.869 ± 0.103 |
| Statistical | SVM-RBF | 0.985 ± 0.020 | 0.836 ± 0.174 |
| FFT | MLP | **1.000 ± 0.000** | **0.971 ± 0.043** |
| FFT | Random Forest | **1.000 ± 0.000** | 0.956 ± 0.054 |
| FFT | SVM-RBF | 0.999 ± 0.003 | 0.774 ± 0.061 |
| Wavelet db4 | MLP | 0.969 ± 0.017 | 0.830 ± 0.087 |
| Wavelet db4 | Random Forest | 0.982 ± 0.011 | 0.892 ± 0.045 |
| Wavelet db4 | SVM-RBF | 0.929 ± 0.018 | 0.850 ± 0.074 |
| STFT spectrogram | MLP | 0.998 ± 0.003 | 0.916 ± 0.056 |
| STFT spectrogram | Random Forest | 0.980 ± 0.011 | 0.898 ± 0.069 |
| STFT spectrogram | SVM-RBF | 0.999 ± 0.001 | 0.942 ± 0.054 |
| *Descriptive reference, not compared* | | | |
| CNN 2D spectrogram | CNN2D | 0.971 ± 0.116 | 0.957 ± 0.119 |

### 3.2. Cross-load degradation

Figure 3 shows macro-F1 for every (training load, test load) pair, one panel per representation, diagonal boxed. Averaged over the twelve off-diagonal cells, the drop from the diagonal is 0.0740 for the spectrogram, 0.0995 for FFT magnitude, 0.1029 for the wavelet and 0.1534 for time-domain statistics. These are numerical differences. Resampling the twelve pairs gives intervals of 0.043–0.105, 0.080–0.122, 0.073–0.133 and 0.074–0.240, approximate because the pairs share loads; the primary test reported below does not separate the four representations, and leaving one load out moves the drops within 0.052–0.096, 0.074–0.113, 0.067–0.132 and 0.024–0.207, reversing the position of time-domain statistics. No ordering is statistically supported, and the smaller drop of the spectrogram is not evidence that it is intrinsically more robust to load changes. Descriptively, the large drop of time-domain statistics comes from transfers involving 0 HP: without that load it is 0.024, the smallest of the four.

The matrix is also asymmetric. With time-domain statistics, training at 0 HP and testing at 1 HP gives 0.795 against 0.678 for the reverse; at maximum separation the two directions give 0.771 and 0.539, the worst cell of the matrix. With statistics and the wavelet, testing at 0 HP is harder in every pair, and the healthy class has 233 windows there against about 468 elsewhere, while FFT magnitude shows the opposite asymmetry. The healthy class does not account for all of it. Averaged over pairs, macro-F1 transferring towards the lower load minus towards the higher is −0.082 for time-domain statistics and −0.063 with the healthy class excluded, and the other three representations also keep their sign without it.

The convolutional network is a descriptive reference only (Table 3): it uses a different input and architecture and enters no comparison. Trained at 3 HP it failed in one of the five folds, whose four cells fall between 0.396 and 0.489 and produce its standard deviations of 0.116 and 0.119; it is reported as it ran, since the epoch budget was fixed in advance.

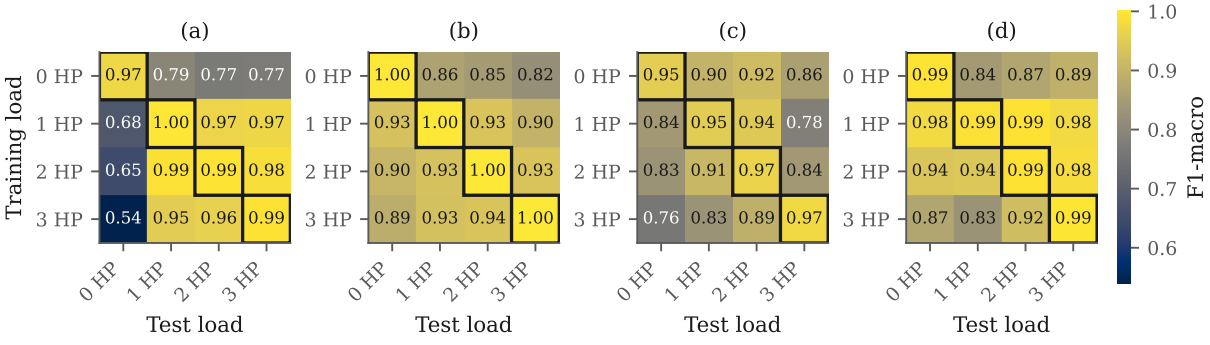



**Figure 3.** Macro-F1 for every (training load, test load) pair, averaged over the three vector classifiers and 5 folds. Rows are the training load and columns the test load. (a) Statistical, (b) FFT, (c) Wavelet db4, (d) STFT spectrogram. The diagonal is a reference cell, boxed for orientation, and takes part in no test.

### 3.3. No consistent ranking across classifiers

With the default settings of the protocol, the primary Friedman test over the twelve off-diagonal pairs does not reject ($\chi^2 = 5.20$, $p = 0.158$; mean ranks 1.833 for the spectrogram, 2.500 for FFT magnitude, 2.667 for time-domain statistics and 3.000 for the wavelet, against a Nemenyi critical distance of 1.354), and no pair is separated in Figure 4, the smallest post-hoc $p$-value being 0.119.

The three per-classifier tests each reject at the Bonferroni-corrected $\alpha = 0.0167$ and disagree on the ordering. The spectrogram ranks first with SVM-RBF ($\chi^2 = 17.10$, $p = 0.00067$), where FFT magnitude ranks last at 3.500, while FFT magnitude ranks first with Random Forest ($\chi^2 = 13.70$, $p = 0.0033$, rank

1.333) and the perceptron ($\chi^2 = 21.00$, $p = 0.00011$, rank 1.083). Averaging them can offset that disagreement, which is consistent with the omnibus test not rejecting while its components do.

The fold-level ANOVA of the protocol is not interpreted. On the 48 cell means only load distance remains ($F = 5.85$, $p = 0.006$); representation gives $F = 2.62$, $p = 0.065$, and its interaction with load distance $F = 0.85$, $p = 0.543$. Adding the classifier locates the effect elsewhere (Table 4): in these configurations the representation by classifier interaction is the largest term ($F = 4.91$, $p < 0.001$, $\eta_p^2 = 0.214$), while representation by load distance is not significant ($F = 1.76$, $p = 0.113$). With the pair as a block the interaction stays at $F = 5.78$ ($p < 0.001$) and the block has the largest partial effect size ($\eta_p^2 = 0.307$) in this exploratory model.

The analyses added for the dependence between pairs qualify these results without overturning them. In the jackknife the interaction remains in all four reduced sets ($F$ from 2.72 to 11.66, $p \leq 0.022$) and each per-classifier leader is unchanged (once tied), although with $N = 6$ eight of the twelve tests fall short of the corrected threshold (Kendall's $W$ from 0.30 to 0.94). The primary ordering is less stable: without 0 HP time-domain statistics rank first and the test rejects ($p = 0.006$), and without any other load they rank last. The mixed model could be fitted only without the classifier by load terms, and a quarter or more of its bootstrap refits failed, so its likelihood ratios, 2.29 for representation and 21.74 for the interaction (indicative $p = 0.61$ and 0.005), are not relied on.

Nested tuning changes the rankings but not the disagreement. The per-classifier tests still reject with the same leaders, the blocked interaction stays at $F = 4.66$ ($p < 0.001$), and the primary test now rejects ($\chi^2 = 9.30$, $p = 0.026$; mean ranks 1.92 for FFT magnitude, 2.17 for the spectrogram, 2.50 for the statistics and 3.42 for the wavelet). The change comes mostly from the SVM: tuning raised its cross-load macro-F1 with FFT magnitude from 0.774 to 0.833, out of last place, and with the wavelet raised in-domain macro-F1 from 0.929 to 0.975 while cutting cross-load macro-F1 from 0.850 to 0.680, consistent with an inner validation that rewards fit to the training load. The forest kept its defaults in 69 of 80 models. The rankings are therefore specific to the configurations evaluated, not the best each representation can reach, and whether the primary test rejects depends on them.

**Table 4.** Exploratory three-way ANOVA on macro-F1 over the 144 off-diagonal cell means, with default hyperparameters; $p$-values are not adjusted for the dependence between cells.

| Source | SS | df | $F$ | $p$ | $\eta_p^2$ |
|---|---|---|---|---|---|
| Representation (R) | 0.1595 | 3 | 5.47 | 0.002 | 0.132 |
| Classifier (C) | 0.0687 | 2 | 3.53 | 0.033 | 0.061 |
| Load distance (D) | 0.2370 | 2 | 12.18 | $< 0.001$ | 0.184 |
| R × C | 0.2865 | 6 | 4.91 | $< 0.001$ | 0.214 |
| R × D | 0.1030 | 6 | 1.76 | 0.113 | 0.089 |
| C × D | 0.0178 | 4 | 0.46 | 0.766 | 0.017 |
| R × C × D | 0.0336 | 12 | 0.29 | 0.990 | 0.031 |
| Residual | 1.0504 | 108 | | | |

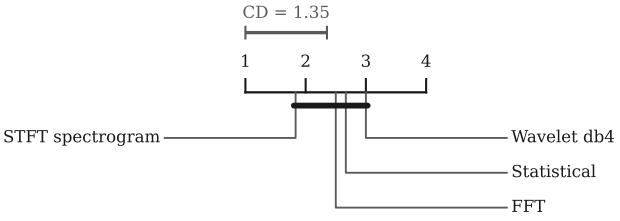



**Figure 4.** Nemenyi critical-difference diagram for the primary Friedman test ($N = 12$ off-diagonal pairs, $k = 4$ representations, $\chi^2 = 5.20$, $p = 0.158$). Representations joined by a bar are not distinguishable at $\alpha = 0.05$; CD is the Nemenyi critical distance.

### 3.4. Performance by fault diameter

Over the off-diagonal cells recall is highest for the healthy class (0.994) and lowest for inner race faults at 0.014 in (0.587). The next four subgroups are ball at 0.021 in (0.786), inner race at 0.021 in (0.823), inner race at 0.007 in (0.832) and outer race at 0.014 in (0.849); the remaining four sit above 0.89. For inner race faults at 0.014 in recall is 0.836 with FFT magnitude against 0.485 with the spectrogram, 0.494 with time-domain statistics and 0.533 with the wavelet. These figures describe where classification fails across loads, not why: label verification found weak signatures in two of the weaker subgroups but not in the weakest, and nothing here identifies a physical cause. The subgroup analysis is descriptive and secondary to the question of the study.

### 3.5. Amplitude normalisation in the tested configurations

The hypothesis behind the ablation is that representations retaining absolute amplitude degrade because amplitude scales with load. Dividing each window by its RMS value before extracting time-domain statistics and FFT magnitude tests that procedure with SVM-RBF and Random Forest. No combination improved by an interval that excludes zero, and two de-

graded. With Random Forest, time-domain statistics lost 0.149 macro-F1 (95 % CI −0.197 to −0.103, Cliff's delta −0.472) and FFT magnitude lost 0.133 (95 % CI −0.151 to −0.116, delta −0.750). With SVM-RBF both intervals cross zero: −0.018 for statistics (95 % CI −0.047 to +0.011, delta −0.205) and +0.033 for FFT (95 % CI −0.002 to +0.067, delta 0.330). The result separates the two classifiers rather than ruling out any benefit. It does not show that amplitude normalisation is ineffective in general, only that this procedure gave no consistent improvement here; on this dataset, removing amplitude this way is inadvisable with the forest and of uncertain effect with the SVM.

### 3.6. Limitations

Each combination of fault type and diameter in the CWRU dataset is a single physical bearing measured under four loads, so crossing loads keeps the bearing fixed and a model may rely on the signature of that specimen rather than on the fault type. The results measure transfer between loads of the same specimens, not generalisation to other bearings or machines. For that purpose they are likely optimistic, since specimen-specific cues may survive a change of load, and they should not be carried to industrial settings without external validation.

The diagonal cells cannot use a record-level split, since the dataset provides one healthy record per load, so the in-domain figures of Table 3 are a reference level, not an estimate of generalisation. The same record leaves the healthy class imbalanced differently across loads, 233 windows at 0 HP against about 468 elsewhere; macro-F1 and class weighting mitigate this without removing the influence of the dataset structure, and the perceptron accepts no class weights. The drop of time-domain statistics is concentrated in transfers involving 0 HP, although the direction asymmetry is not confined to the healthy class.

The envelope spectrum, used here to verify the labels, is not among the four representations, which limits what the comparison covers.

Tuning used only the training load, so the tuned results are not the best transfer each representation can reach, and the grids are small. The four representations also differ in dimension from 6 to 1025, which the default $\gamma$ of the SVM ties to the kernel width; dimension was not equalised because it is part of a representation.

With four loads every inference rests on few independent units: the mixed model could not be fitted in full, its bootstrap was unstable, and the jackknife leaves six pairs. The $p$-value of 0.158 is a failure to separate the four representations, not evidence that they are equivalent.

Four limitations therefore remain unresolved: generalisation beyond one specimen per fault condition, which bounds every result; inference with four loads, which makes all $p$-values indicative; the dependence of the rankings on the configuration, which restricts them to the settings evaluated; and the cause of the weak subgroups, which these data do not identify.

## 4. Conclusions

Within a single load the four representations average above 0.95 macro-F1 and separate by less than four hundredths, so on this dataset a comparison under the training condition alone does not identify a preferable one. Across loads the mean drops range from 0.0740 for the STFT spectrogram to 0.1534 for time-domain statistics, but the primary test does not separate the representations and the largest depends on 0 HP, so no ordering is established.

No representation was superior across the classifiers and configurations evaluated. With default hyperparameters the omnibus Friedman test does not reject, which is not evidence of equivalence, while the per-classifier tests reject and place different representations first, FFT magnitude ranking last with SVM-RBF and first with the other two. The representation by classifier interaction persists when any load is left out and under nested tuning, which however changes the rankings and makes the omnibus test reject, so it is exploratory evidence specific to these configurations: on this dataset a representation is best assessed with the classifier it will be used with.

Per-window RMS normalisation gave no clear improvement in cross-load transfer in the four tested combinations and degraded it with Random Forest, by 0.149 for time-domain statistics and 0.133 for FFT magnitude.

All of this concerns transfer between loads of the same bearings. External validation on other bearings and machines is needed before industrial use.

## Data availability

The CWRU records are public [2]. The repository, withheld for double-blind review and provided upon acceptance, holds the per-fold results and the code of every step, from download, label verification, partition and leakage check to feature extraction, classifier configuration, metrics, statistics, nested tuning and figures. One command reruns the pipeline from the raw records, whose download by the first script is the only external step, and regenerates every table and figure. On CPU the results are identical with the fixed seed, although the perceptron on low-dimensional inputs also depends on the number of BLAS threads (the pipeline uses the default), and the convolutional network reproduces on the original GPU. A full rerun

from a fresh copy of the repository regenerated every table identically, apart from timing columns.

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